

# Raushan Kumar — Professional Knowledge Base

This document contains structured information about Raushan Kumar, a Full Stack AI Engineer. It is organized into self-contained sections so that each section can be retrieved and understood independently by a Retrieval-Augmented Generation (RAG) system.

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## Basic Contact Information

Raushan Kumar is a Full Stack AI Engineer based in Ranchi, Jharkhand, India. His email address is [raushankumar225024@gmail.com](mailto:raushankumar225024@gmail.com) and his phone number is 8709309119. His LinkedIn profile is available at <https://www.linkedin.com/in/raushan-kumar-748837259>, and his GitHub username is Raushan225024, accessible at <https://github.com/Raushan225024>.

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## Professional Summary

Raushan Kumar is a Full Stack AI Engineer who specializes in integrating Artificial Intelligence into web applications and IoT systems. He builds full-stack applications and AI-powered agents from scratch, combining modern web technologies with Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and intelligent automation.

Raushan's main areas of expertise include MERN Stack development (MongoDB, Express.js, React.js, Node.js), Next.js and full-stack web application development, SQL and database management, AI agent development using LangChain, Retrieval-Augmented Generation (RAG) systems, LLM integration and AI application development, vector databases and semantic search, API development and AI system integration, and IoT and AI integration.

Raushan is particularly interested in Artificial Intelligence, Machine Learning, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents and Agentic Systems, Backend Development, Full Stack Development, AI-Powered Web Applications, and IoT and Intelligent Systems.

Raushan's professional goal is to build scalable, reliable, and intelligent software solutions that solve real-world problems by combining modern web development, Artificial Intelligence, AI agents, and intelligent automation.

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## Primary Role and Areas of Specialization

Raushan Kumar's primary professional role is Full Stack AI Engineer. He specializes in four main areas: the MERN Stack, Artificial Intelligence, LLM Applications, and RAG Systems.

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## Experience Level

Raushan Kumar is currently a fresher with zero years of formal professional experience, though he has hands-on experience through multiple personal and academic projects. He is capable of working on backend API development, AI-powered applications, frontend UI development, RAG applications, AI search systems, database design, vector database integration, LLM integration, and automation systems.

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## Technical Skills: Programming Languages

Raushan Kumar is proficient in Python, JavaScript, and C++.

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## Technical Skills: Backend Development

Raushan Kumar's backend development skills include FastAPI, Flask, Django, Node.js, Express.js, REST API development, API integration, and Redis.

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## Technical Skills: Frontend Development

Raushan Kumar's frontend development skills include React.js, Next.js, HTML, CSS, Tailwind CSS, and Redux Toolkit.

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## Technical Skills: Artificial Intelligence and Machine Learning

Raushan Kumar's AI and Machine Learning skills include Machine Learning, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Prompt Engineering, Embeddings, Vector Search, and Retrieval-Augmented Generation (RAG).

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## **Technical Skills: AI Frameworks**

Raushan Kumar has experience with the AI frameworks LangChain and Hugging Face.

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## **Technical Skills: Databases**

Raushan Kumar's relational database skills include PostgreSQL, Supabase, MySQL, and MongoDB.

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## **Technical Skills: Vector Databases**

Raushan Kumar has experience with the vector databases Supabase pgvector, Pinecone, ChromaDB, and FAISS.

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## **Technical Skills: Cloud and Deployment**

Raushan Kumar's cloud and deployment experience includes Docker, AWS, Azure, and Vercel.

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## **Technical Skills: Development Tools**

Raushan Kumar regularly uses Git, GitHub, VS Code, and Postman as part of his development workflow.

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## **Project 1: Smart Locker Booking Application**

Raushan Kumar built a Smart Locker Booking application that allows users to book a physical locker to store belongings such as bags, purses, phones, and other items for a fixed period of time. During booking, the user sets a password, and the corresponding locker can later be opened using either the user's phone or the locker's own dashboard.

**Problem this project solves:** Tourists and travelers often struggle to safely store their personal belongings while visiting tourist destinations, monuments, religious places, and museums, many of which do not allow visitors to carry phones, electronics, backpacks, or large luggage inside. Travelers are often forced to leave belongings with unauthorized individuals or unreliable storage services, creating risks of theft, loss, or damage. This project addresses the lack of a reliable, standardized, and easily accessible luggage storage network by providing a Smart Locker Booking and Storage Network where users can find nearby lockers, check real-time availability, book a locker in advance, make payments, and securely access their belongings using technologies such as OTP, QR codes, digital authentication, and real-time monitoring.

**Technologies used in the Locker Booking project:** JavaScript, C++, React.js, Express.js, MongoDB, Redux Toolkit, REST APIs, WebSocket, payment integration, and Tailwind CSS.

**System architecture:** User → Frontend/API → Backend → Database/Vector Database → Response.

**Raushan's responsibilities on this project** included designing the system architecture, developing backend APIs, database integration, AI model integration, testing, and deployment.

#### **Key features of the Locker Booking application:**

1. Online locker booking without needing to physically visit the location.
2. Real-time locker availability status.
3. UPI payment integration for secure payments.
4. Locker booking management from the user's profile.
5. User profile management.
6. Secure authentication system for registration and login.
7. Password management for changing account passwords.
8. Booking history showing previous and current bookings.
9. A user-friendly web application interface for searching, booking, paying for, and managing lockers.
10. A secure and convenient storage solution that reduces the risk of theft, loss, or unauthorized handling of belongings.

**GitHub repositories for the Locker Booking project:** Frontend code is at <https://github.com/Raushan225024/bookingappui.git> and backend code is at <https://github.com/Raushan225024/bookingapp.git>. A live demo is available at <https://bookingappui-8fgf.vercel.app/>.

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## Project 2: "Chat About Me" — Personal RAG-Based AI Agent

Raushan Kumar built a Personal RAG-Based AI Agent called "Chat About Me," an intelligent question-answering system designed to provide accurate information about his own professional profile. The system allows users, recruiters, and clients to ask natural-language questions about Raushan's skills, projects, education, experience, and technical expertise.

The project uses Retrieval-Augmented Generation (RAG) technology to retrieve relevant information from a personal knowledge base before generating a response. Raushan's personal and professional information is converted into vector embeddings and stored in a vector database. When a user asks a question, the system retrieves the most relevant information and provides it as context to a Large Language Model (LLM) through an API, which then generates a context-aware and accurate response. This approach reduces irrelevant responses and lets the AI agent act as a virtual AI representative for recruiters and clients.

**Problem this project solves:** Normally, recruiters or clients need to review multiple sources such as resumes, portfolios, and project descriptions to understand a person's skills, experience, education, and projects, which can be time-consuming. This RAG-based agent solves that problem by creating an interactive AI system that can answer questions about Raushan's professional profile in real time, without requiring manual searching through multiple documents.

**Technologies used in the "Chat About Me" project:** Python as the programming language; Groq API and the openai/gpt-oss-20b model for the LLM; LangChain as the RAG framework; Hugging Face Embeddings and Sentence Transformer models for embeddings; Python for the backend; JavaScript, React, and Redux Toolkit for the frontend; and python-dotenv for environment variable configuration. Core AI techniques used include Retrieval-Augmented Generation (RAG), vector embeddings, semantic search, similarity search, and Natural Language Processing (NLP).

**Raushan's responsibility on this project:** He was responsible for building the entire system.

### Key features of the "Chat About Me" AI agent:

1. Personal question answering about skills, projects, education, and experience.
2. A RAG-based architecture that retrieves relevant information before generating a response.
3. Context-aware responses grounded in the personal knowledge base.
4. Semantic search using vector embeddings to understand query meaning.
5. Vector database integration for efficient similarity search.
6. LLM API integration for generating natural, human-like responses.
7. A personalized knowledge base containing Raushan's skills, projects, experience, and background.

8. Natural language interaction without requiring specific keywords or commands.
9. Relevant information retrieval before sending context to the LLM.
10. Acting as an AI-powered professional assistant / virtual representative for recruiters and clients.

**GitHub repositories for the "Chat About Me" project:** Backend code is at [https://github.com/Raushan225024/rag\\_3rd\\_project.git](https://github.com/Raushan225024/rag_3rd_project.git) and frontend code is at [https://github.com/Raushan225024/ai\\_mern.git](https://github.com/Raushan225024/ai_mern.git).

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## Project 3: AI-Powered Relational Database Semantic Search Pipeline

Raushan Kumar built an AI-Powered Relational Database Semantic Search Pipeline designed to enable intelligent, meaning-based search over data stored in traditional relational databases. Traditional relational databases are efficient for structured, exact-match SQL queries, but struggle with searches based on the meaning or intent of a user's query. This project addresses that limitation with an automated pipeline that extracts structured records from a relational database, converts them into human-readable text while preserving each record's unique identifier, and then converts that text into high-dimensional vector embeddings using an AI embedding model. These embeddings, which capture the semantic meaning of each record, are stored in a vector database alongside their corresponding unique IDs. When a user submits a natural-language search query, the system converts it into a vector embedding using the same embedding model, performs a semantic similarity search against the stored vectors, and returns the unique IDs of the most relevant records, which can then be used to retrieve the complete, accurate information from the original relational database.

**Problem this project solves:** Traditional relational databases primarily support structured, keyword-based queries, so users often need to know exact keywords or field names to retrieve relevant information. For example, a search for "comfortable furniture for a small living room" may return no results from a keyword-based SQL search if the database doesn't literally contain those words. This project addresses limited semantic understanding in SQL search, dependence on exact keyword matching, difficulty searching large datasets using natural language, and the challenge of integrating AI-powered search with existing relational databases.

**Proposed solution:** The pipeline works alongside the existing relational database rather than replacing it. Each database record is extracted, converted into human-readable text, converted into a vector embedding via an AI embedding model, and stored in a vector database with its unique record ID. At search time, the user's natural-language query is converted into a query embedding, compared against stored vectors via semantic similarity search, and the unique IDs of the most relevant records are used to retrieve complete data from the original relational database.

**System architecture (data ingestion path):** Relational Database → Data Extraction → Convert Records into Readable Text → AI Embedding Model → Generate Embeddings → Vector Database plus Unique Record ID.

**System architecture (query path):** User Query → Query Embedding → Semantic Similarity Search → Relevant Unique Record IDs → Retrieve Complete Record Data → Search Results.

**Worked example of the pipeline:** A database record such as "Product ID: 101, Product Name: Modern Wooden Sofa, Category: Furniture, Description: Comfortable sofa suitable for living rooms" is transformed into the readable text "Product ID 101. Modern Wooden Sofa is a furniture product. It is a comfortable sofa suitable for living rooms." This text is passed through an embedding model to produce a vector embedding (e.g., [-0.038, 0.025, 0.091, ...]), which is stored together with its record ID (e.g., {"record\_id": 101, "embedding": [vector values]}). When a user later searches "Comfortable furniture for my living room," the query is embedded and compared against stored vectors, and the system returns the IDs of the most semantically relevant records (e.g., 101, 205, 342), which are used to fetch the complete data from the relational database.

**Key features of the Semantic Search Pipeline project:**

1. Relational database integration for extracting structured records.
2. An automated data processing pipeline that prepares records for AI-powered semantic search.
3. Conversion of structured data into readable text before embedding.
4. AI-based vector embeddings using an embedding model.
5. Efficient vector database storage of embeddings.
6. Unique ID mapping between each embedding and its original database record.
7. Semantic search using natural language rather than exact keywords.
8. Similarity-based retrieval of relevant records.
9. Integration with existing databases, which remain the primary source of complete data.
10. A scalable AI search architecture capable of processing large numbers of records.

**Technologies used in the Semantic Search Pipeline project:** Python as the programming language; PostgreSQL/SQL as the relational database; Supabase, PostgreSQL, and pgvector as the vector database layer; Hugging Face Embeddings and Sentence Transformer models for AI/ML; and vector embeddings, semantic search, similarity search, and NLP as the core AI techniques, built on a Python-based data processing pipeline.

**Expected benefits of this project:** It enables natural language search over relational database records, reduces dependency on exact keyword matching, improves search relevance through semantic similarity, preserves the existing relational database infrastructure, bridges traditional databases and AI-powered search, and supports efficient retrieval using unique record identifiers.

**Applications of this project** include e-commerce product search, enterprise database search, customer information retrieval, inventory search systems, job and candidate search, real estate search platforms, educational data retrieval, healthcare information systems, and intelligent recommendation systems.

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## **Current Research Project: Resource-Adaptive Lightweight Object Detection for Real-Time Autonomous Edge Systems**

Raushan Kumar is currently working on a research project titled "Resource-Adaptive Lightweight Object Detection for Real-Time Autonomous Edge Systems." Autonomous mobile systems — including autonomous robots, unmanned aerial vehicles (UAVs/drones), and autonomous vehicles — are increasingly deployed in applications requiring continuous, real-time perception of their surrounding environment. Object detection, the task of localizing and classifying objects within a visual scene, is a foundational component of the perception pipeline in such systems, directly supporting downstream decision-making, navigation, and control.

While deep learning-based object detection models can achieve high accuracy on benchmark datasets, they are typically computationally intensive, and deploying them directly on edge devices — the embedded or onboard computing units carried by robots, drones, and vehicles — is constrained by limited processing power, restricted memory, and finite energy or battery capacity. Edge devices must also satisfy strict real-time latency requirements, since delayed detection can compromise safety and operational reliability.

This research is situated within the broader field of Edge Artificial Intelligence (Edge AI), which investigates how machine learning workloads can be efficiently executed on resource-constrained hardware close to the point of data acquisition rather than on remote cloud servers. The specific focus of Raushan's proposed work is real-time object detection for autonomous edge systems, with an emphasis on adapting to the resource conditions that are actually available on the device during operation.

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## **RAG (Retrieval-Augmented Generation) Experience and Understanding**

Raushan Kumar has practical experience and knowledge in building Retrieval-Augmented Generation (RAG) systems. His understanding spans the full RAG pipeline, including document ingestion, data cleaning, text processing, chunking strategies, embedding generation, vector

databases, semantic search, metadata filtering, retrieval, context generation, and LLM response generation.

**Raushan's preferred RAG architecture for document ingestion:** Documents → Data Cleaning → Text Processing → Chunking → Embedding Model → Vector Database.

**Raushan's preferred RAG architecture for answering a user question:** User Question → Query Embedding → Vector Search → Relevant Documents → LLM → Final Answer.

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## Technical Knowledge: Python

Raushan Kumar's Python knowledge includes functions, object-oriented programming, modules, packages, virtual environments, error handling, API development, and database integration.

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## Technical Knowledge: Databases

Raushan Kumar's database knowledge includes SQL, PostgreSQL, database schema design, CRUD operations, joins, indexes, and vector search.

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## Technical Knowledge: Artificial Intelligence

Raushan Kumar's AI knowledge includes embeddings, vector representations, semantic search, transformers, Large Language Models, and Retrieval-Augmented Generation.

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## Services Raushan Kumar Can Provide

**Backend development services:** REST API development, Python backend development, database integration, and API integration.

**AI development services:** AI-powered applications, LLM integration, chatbots, RAG systems, and AI search systems.

**Database development services:** PostgreSQL database design, Supabase integration, and vector database implementation.

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## Preferred Technology Stack

Raushan Kumar's preferred backend technologies are Python, REST APIs, and FastAPI. His preferred AI frameworks are LangChain and Hugging Face. His preferred databases are PostgreSQL, Supabase, and MongoDB. His preferred vector search technology is pgvector. His preferred deployment tools are Docker and Railway.

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## Working Style

Raushan Kumar's working approach includes understanding requirements before development, designing a clear system architecture, writing maintainable code, testing functionality, using version control, and documenting important technical decisions.

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## Professional Strengths

Raushan Kumar's professional strengths include problem-solving, learning new technologies quickly, backend development, AI application development, database design, and system architecture.

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## Technologies Raushan Kumar Is Currently Learning

Raushan Kumar is currently learning Deep Learning, PyTorch, Transformer architecture, AI model training, advanced RAG systems, and AI agents.

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## Career Objectives

Raushan Kumar's professional goal is to become an AI Engineer, Machine Learning Engineer, or Senior Software Engineer. He is particularly interested in working on Artificial Intelligence products, Machine Learning applications, AI Agents, LLM applications, and large-scale backend systems.

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## Availability

Raushan Kumar is available for full-time opportunities, freelance projects, contract work, and collaboration. He is open to remote, hybrid, or on-site work arrangements.

**How to contact Raushan Kumar:** Email [raushankumar225024@gmail.com](mailto:raushankumar225024@gmail.com), connect on LinkedIn at <https://www.linkedin.com/in/raushan-kumar-748837259>, or view his work on GitHub under the username Raushan225024.

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## Educational Qualifications: Integrated B.Tech & M.Tech

Raushan Kumar is pursuing an Integrated B.Tech and M.Tech degree, specializing in Machine Learning and Data Science, at the Central University of Jharkhand. His program duration is from 2022 to 2027, and he is currently in his 9th semester with a CGPA of 7.64. Relevant areas of study include Artificial Intelligence, Machine Learning, Data Science, and related technologies.

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## Educational Qualifications: Senior Secondary Education (Class XII)

Raushan Kumar completed his Class XII (Senior Secondary) education under the Bihar School Examination Board at Anurgh Narayan College, Gaya, in 2020, scoring 65.5 percent.

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## Educational Qualifications: Secondary Education (Class X)

Raushan Kumar completed his Class X (Secondary) education under the Bihar School Examination Board at Prakash Vidya Mandir, Tikari, in 2018, scoring 66.4 percent.